

One Hundred Publication-Quality Figures

A practical guide, with the complete worked-example library

Yanwei Wang

companion to the `publication-figures` repository (scripts `fig001-fig100+`, `publication.mplstyle`, `journal_style.py`, `ashby.py`)

Contents

1	The craft: how these figures are made	1
1.1	One style sheet for the whole paper	1
1.2	Derive, don't draw	1
1.3	Geometry and typography	1
1.4	Colour and encoding	2
1.5	Annotation beats legends	2
1.6	Multi-panel discipline	2
1.7	Verification is part of the figure	2
1.8	Reproducibility and packaging	2
2	The catalog: one hundred worked examples	2
3	The failure-mode catalog	13
4	Checklists	14

1 The craft: how these figures are made

Every figure in this library was produced the same way, and the method is the point. The one hundred scripts are worked examples of eight habits.

1.1 One style sheet for the whole paper

All scripts load a single `publication.mplstyle`: single-column figures about 3.5–4.0in wide (double column 7.0–7.2in), 7–8pt text so labels match journal captions after scaling, the Okabe–Ito colour-blind-safe cycle, despined axes with outward ticks (the `journal-style` module switches to boxed inward ticks when imitating older journals), `constrained.layout` for spacing, and twin outputs: a 600-dpi PNG for previewing and a vector PDF for submission. Consistency is not cosmetic — reviewers read a paper whose figures share one visual grammar far faster.

1.2 Derive, don't draw

No curve in this library is sketched. Each is computed from the governing equations — Nernst lines, Butler–Volmer kinetics, the Buckley–Leverett fractional flow, Planck's law — with generic textbook parameters. When the task is to *reproduce* a figure from a paper (fig. 61), the workflow is: read every label and axis; derive the underlying solution; verify landmark values the original states; parameterise the computation; add asymptotic expansions the original implies; imitate the styling (“style archaeology”); iterate against the target. A reproduced figure you derived is one you understand — and one you may legally redraw, because the physics is nobody's property while the original's pixels are.

1.3 Geometry and typography

Work in inches and points, never pixels: set the final printed size and let DPI follow. Label every axis with symbol *and* unit. Prefer `mathtext` for symbols and remember it is a dialect of $\text{\LaTeX}$, not $\text{\LaTeX}$ itself (pattern 14). Type μ , $^\circ$ and the minus sign as literal characters — escape sequences inside raw strings print themselves (pattern 1).

1.4 Colour and encoding

Default to a colour-blind-safe cycle and never let colour carry meaning alone: pair it with line style or marker shape, and check the figure in grayscale. Reserve semantic colour (red for failure envelopes, warm for hot isotherms) for when it teaches.

1.5 Annotation beats legends

With few curves, label them directly, in their own colour, where the curves are *separated* — converging families (sag curves, titration tails) must be labelled in their fan, not at their confluence (pattern 11). Rotated labels along sloped lines must use the rendered axes aspect, not the data slope (pattern 2); `ashby.slope_angle` automates this. Keep annotations off the data; use thin leaders when the target region is crowded; aim leaders at the named object's ink, not the nearest convenient coordinate (pattern 15). Bottom-edge labels collide with tick numbers — place them inside the axes (pattern 10).

1.6 Multi-panel discipline

One message per panel; bold sequential panel letters; shared axes where comparison is the point; `height_ratios` for sketch strips and residual panels; insets never over data — relocate or promote to a panel (pattern 3).

1.7 Verification is part of the figure

Run, *look at the pixels*, fix, rerun — plausible code routinely produces wrong figures, and the render is the only witness. Build self-checks into the physics wherever possible: the Wien locus must thread every Planck peak (fig. 93); Mohr circles must touch the envelope because centre-to-line distance equals radius (fig. 85); the Carnot loop integral must equal the T - s rectangle, and an `assert` says so (fig. 100). Seed every random generator; print confirmations and key derived numbers on save.

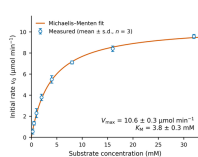
1.8 Reproducibility and packaging

One standalone script per figure with a marked DATA block; shared code promoted to modules (`journal_style.py`, `ashby.py`); a README indexing every script in one line; the whole library shipped as a versioned zip. A figure others can regenerate is a figure others can trust.

2 The catalog: one hundred worked examples

Each entry shows the figure, what it demonstrates, and (where the description names one) the built-in check it carries. Thumbnails are downsampled; run the script for the full-resolution PNG and vector PDF.

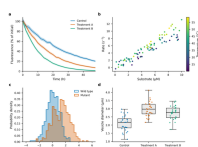
A • Statistical and general data graphics (figs. 1–50)



1. Data + model fit

Data + model fit with error bars (`curve_fit`, parameter annotation)

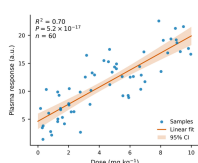
Techniques: annotation



2. Multipanel

2×2 composite with bold a–d panel labels, s.e.m. bands, colorbar, distributions, box + raw points

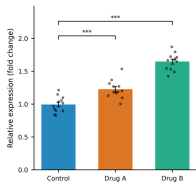
Techniques: colorbar, direct labels



3. Regression ci

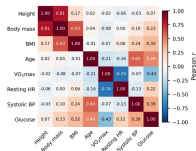
Scatter + linear regression with an analytically computed 95% confidence band and $R^2/P/n$ annotation

Techniques: annotation



4. Bars stats

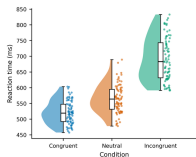
Bars with individual data points, s.e.m. error bars, and significance brackets from t-tests



5. Heatmap

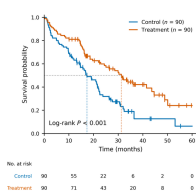
Annotated correlation heatmap with a diverging colormap pinned to [-1, 1]

Techniques: annotation



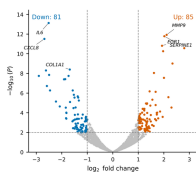
6. Raincloud

Raincloud plots: half-violin distribution + box summary + jittered raw data



7. Kaplan-Meier survival

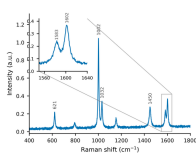
Kaplan-Meier survival curves with censor marks, median guides, log-rank test (`scipy ≥ 1.11`), and a number-at-risk table



8. Volcano

Volcano plot with thresholds, per-side labeled top hits, and rasterized points for small PDFs

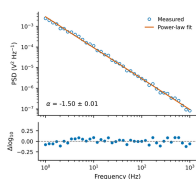
Techniques: rasterized points, direct labels



9. Spectrum inset

Annotated spectrum with a zoom inset via `inset_axes` + `indicate_inset_zoom`

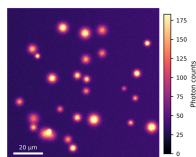
Techniques: inset axes, annotation



10. Loglog powerlaw

Log-log power-law fit with a residual panel (`height_ratios` + `sharex`)

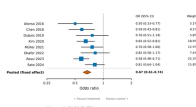
Techniques: log axes



11. Image scalebar

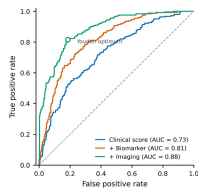
Microscopy-style image: scale bar, percentile contrast stretch, calibrated colorbar, no axis ticks

Techniques: colorbar



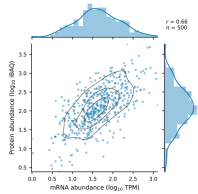
12. Forest plot

Meta-analysis forest plot: weighted squares, pooled diamond, aligned OR/weight text columns, heterogeneity stats



13. ROC curves

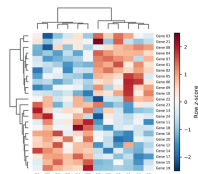
ROC curves with AUC computed from scratch in numpy and a marked Youden optimum



14. Joint marginals

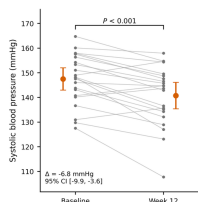
Scatter with marginal histograms/KDEs via GridSpec (a "joint plot" without seaborn)

Techniques: GridSpec



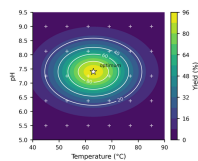
15. Clustered heatmap

Clustered heatmap with scipy dendrograms aligned to the reordered matrix (a "clustermat" from scratch)



16. Paired slope

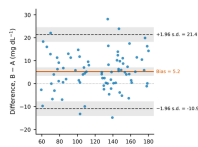
Paired before/after slope chart with per-subject lines, group means $\pm$ 95% CI, and paired t-test



17. Contour map

2-D response surface: filled + labelled contours, design points, marked optimum

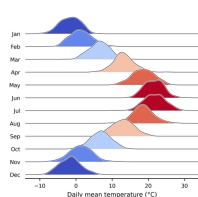
Techniques: direct labels, shading, contours



18. Bland–Altman comparison

Bland–Altman method comparison: bias, limits of agreement, CI bands, right-edge value labels

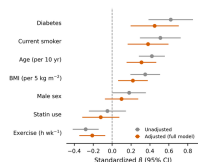
Techniques: direct labels



19. Ridgeline

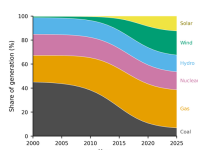
Ridgeline (joy) plot: overlapping KDEs with occlusion and fills colour-mapped to each row's mean

Techniques: shading



20. Coefficients

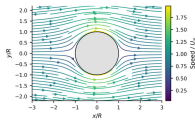
Regression coefficient dot-and-whisker plot comparing adjusted vs unadjusted models



21. Stacked composition

Normalized stacked-area composition with semantic colours and direct right-edge labels (no legend)

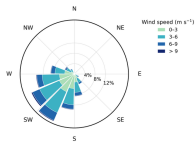
Techniques: legend craft, direct labels



22. Streamplot

Vector field as speed-coloured streamlines with a masked body and an inset colorbar matched to equal-aspect axes

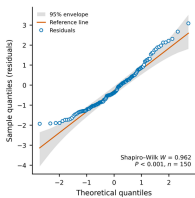
Techniques: inset axes, colorbar, aspect-aware rotation



23. Wind rose

Wind rose: stacked speed classes on a compass-oriented polar axis, radial percent ticks

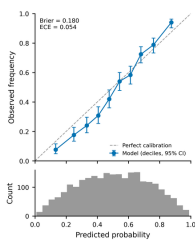
Techniques: polar axes



24. Normal Q-Q plot

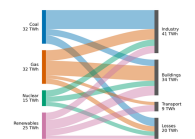
Normal Q-Q plot with quartile reference line, Beta-distribution 95% envelope, and Shapiro-Wilk test

Techniques: envelopes



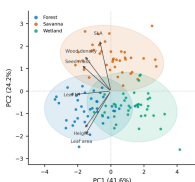
25. Calibration

Calibration curve with Wilson CIs, Brier/ECE, and a prediction-histogram panel



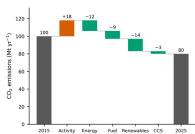
26. Sankey

Sankey/alluvial flow diagram from scratch: smoothstep ribbons via `fill_between`, widths $\propto$ flow



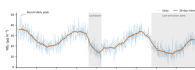
27. PCA biplot

PCA via SVD with 95% confidence ellipses per group and loading arrows



28. Waterfall

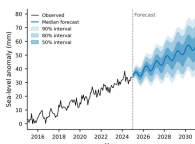
Waterfall chart: floating signed contributions between start and end totals with connectors



29. Time series events

Dated time series: ConciseDateFormatter axis, rolling mean, shaded event periods, annotated peak

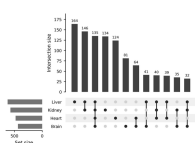
Techniques: series solution, annotation, shading



30. Fan chart

Forecast fan chart: nested 50/80/90% quantile bands from an ensemble with proxy-patch legend

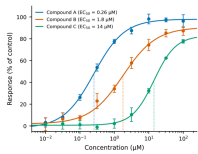
Techniques: legend craft



31. UpSet plot

UpSet plot of set intersections: bars + dot matrix + set sizes, GridSpec with shared axes

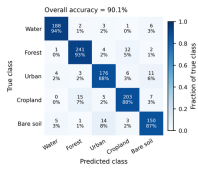
Techniques: GridSpec



32. Dose response

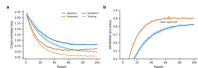
Dose-response curves: 4-parameter logistic fits in log space, EC50 drops and labels

Techniques: direct labels



33. Confusion matrix

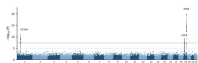
Row-normalized confusion matrix with counts, percentages, and adaptive text colour



34. Training curves

ML training curves: mean $\pm$ s.d. across seeds, train/val line styles, best-epoch marker

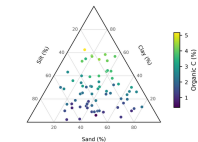
Techniques: seeded RNG



35. Manhattan plot (GWAS)

GWAS Manhattan plot: alternating chromosome colours, significance thresholds, rasterized points, labelled loci

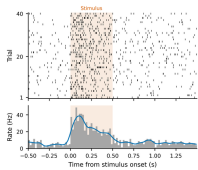
Techniques: rasterized points, direct labels



36. Ternary

Ternary composition diagram from scratch: grid, edge ticks, rotated axis titles, inset colorbar

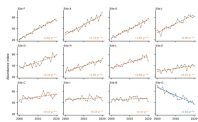
Techniques: inset axes, colorbar



37. Spike raster + PSTH

Spike raster (eventplot) + PSTH with Gaussian-smoothed rate and shaded stimulus window

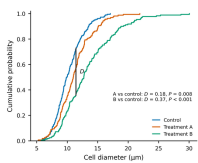
Techniques: shading



38. Small multiples

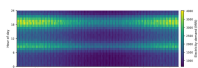
Small-multiples grid: shared axes, per-panel trend fits coloured by sign, sorted panels, supxlabel/supylabel

Techniques: direct labels



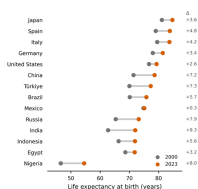
39. ECDF comparison

ECDF comparison via Axes.ecdf with two-sample KS tests and the maximum-gap marker



40. Carpet plot

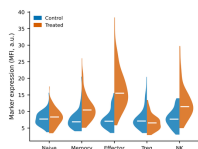
Hour-of-day $\times$ day-of-year carpet plot with pcolormesh cell edges and month-centred ticks



41. Dumbbell

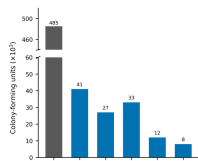
Dumbbell chart of change between two time points, sorted, with Δ labels at the right edge

Techniques: direct labels



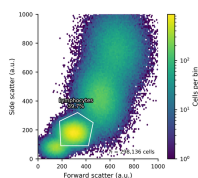
42. Split violin

Split violins: two conditions per category sharing a baseline, white median bars



43. Broken axis

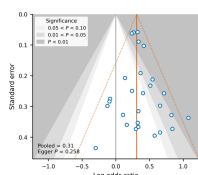
Bar chart with a broken y-axis: stacked axes, hidden joint, diagonal break marks



44. Hexbin density

Hexbin density with LogNorm for $10^5 +$ points, polygon gate with enclosed fraction, stroked labels

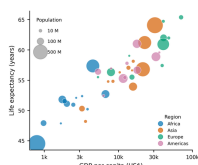
Techniques: direct labels



45. Funnel plot

Contour-enhanced funnel plot: significance shading, pseudo-95% funnel, Egger's test

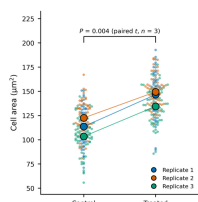
Techniques: shading, contours



46. Bubble chart

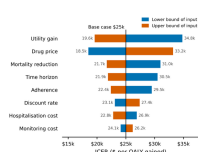
Bubble chart: log axis with plain ticks, colour by group, hand-built size legend (diameter = $\sqrt{\text{area}}$)

Techniques: legend craft



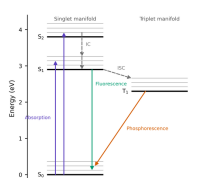
47. Superplot

SuperPlot: from-scratch beeswarm coloured by replicate, replicate means overlaid, test on $n =$ replicates



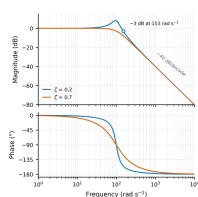
48. Tornado

Tornado diagram for one-way sensitivity analysis, sorted by swing, currency tick formatter



49. Energy levels

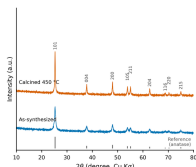
Jablonski energy-level diagram drawn in code: states, vibrational sublevels, radiative/non-radiative arrows



50. Bode plot

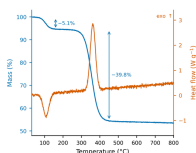
Bode magnitude/phase plot from scipy.signal with -3 dB marker, slope guide, log grid

B • Chemistry and materials characterization (figs. 51–60)



51. X-ray diffraction

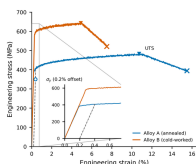
Stacked powder XRD with reference stick pattern and rotated (hkl) indices



52. TGA–DSC thermogram

TGA + DSC on a colour-coded twin axis with mass-loss arrows and "exo up" marker

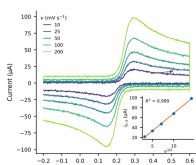
Techniques: twin axes



53. Stress–strain curves

Engineering stress–strain: 0.2% offset yield construction, UTS/fracture markers, elastic-region inset

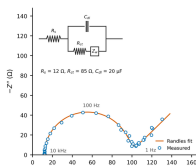
Techniques: inset axes



54. Cyclic voltammetry

CVs at five scan rates with a Randles–Ševčík (i_p vs $\sqrt{v}$) inset

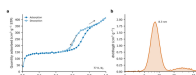
Techniques: inset axes



55. Nyquist EIS

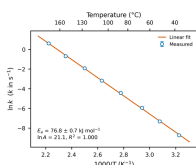
Nyquist impedance plot with Randles-model fit, frequency tags, and a code-drawn equivalent-circuit inset

Techniques: inset axes



56. Isotherm + PSD

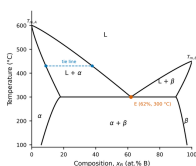
Type-IV N₂ isotherm with H1 hysteresis (IUPAC open/closed symbols) + BJH pore-size panel



57. Arrhenius plot

Arrhenius plot with weighted fit, E_a , and an exact secondary C axis via secondary_xaxis

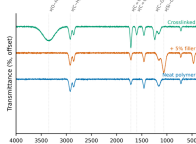
Techniques: secondary axis



58. Phase diagram

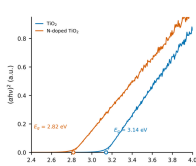
Parameterised binary eutectic phase diagram: labelled fields, eutectic point, tie line

Techniques: direct labels



59. FTIR stack

Stacked FTIR on a reversed wavenumber axis with dotted band guides and rotated assignments

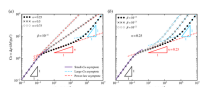


60. Tauc band gap

Tauc plot: linear-region fits extrapolated to the abscissa for optical band gaps

C • Case study: reproducing a journal theory figure (fig. 61)

This single figure carries the reproduction workflow of §I.2 and introduced `journal_style.py` (boxed inward ticks, slope triangles, panel letters), reused by every figure after it.



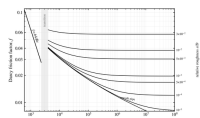
61. Carreau flow curves (case study)

RHEOLOGY

Journal-style theory figure: Carreau slit-flow curves (Cu vs Q) computed from the wall-shear integral, with small- Cu /power-law/large- Cu asymptotes, slope triangles, marker-triple legends, boxed inward-tick axes, and STIX serif math

Techniques: legend craft

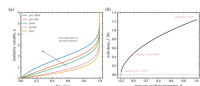
D • Transport phenomena and reactors (figs. 62–67)



62. Moody chart

Moody diagram computed from $64/Re$ and the Colebrook equation (vectorised fixed-point iteration), transition band, direct roughness labels

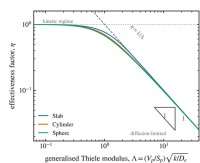
Techniques: direct labels



63. Falkner–Skan profiles

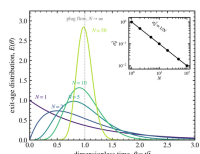
Falkner–Skan boundary layers by shooting (`solve_ivp` + `brentq` with blow-up events): profiles across the pressure-gradient range and wall shear $f''(0)$ vs β with separation/Blasius/stagnation anchors

Techniques: root finding, ODE integration



64. Thiele modulus

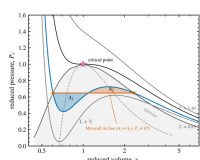
Effectiveness factor vs generalised Thiele modulus for slab/cylinder/sphere ($\tanh$, Bessel I_1/I_0 , $\coth$), asymptotes and slope triangle



65. RTD tanks-in-series

Tanks-in-series residence-time distributions with the $\sigma^2=1/N$ scaling inset

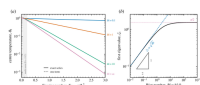
Techniques: inset axes, series solution



66. Van der Waals + Maxwell

Van der Waals isotherms with the Maxwell equal-area construction solved by `brentq` (analytic antiderivative), binodal dome, analytic spinodal, shaded equal-area lobes

Techniques: root finding, shading

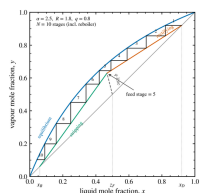


67. Transient conduction

Slab-conduction eigenseries ($\zeta \tan \zeta = Bi$ solved per branch), one-term Heisler approximation with validity band, and ζ_1 vs Bi with both asymptotes

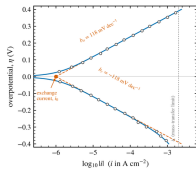
Techniques: series solution

E • Chemical and materials engineering classics (figs. 68–73)



68. McCabe–Thiele construction

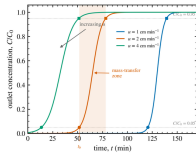
McCabe–Thiele construction with the stage-stepping algorithm coded (equilibrium/operating lines, sloped q -line, numbered stages, automatic feed-stage detection)



69. Tafel plot

Tafel plot from Butler–Volmer kinetics with a cathodic mass-transfer limit, dashed extrapolations meeting at i_0 , slope labels matched to the axes aspect

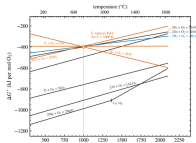
Techniques: direct labels, aspect-aware rotation



70. Breakthrough curves

Fixed-bed breakthrough curves (Thomas-type logistic), breakthrough/exhaustion markers, shaded mass-transfer zone

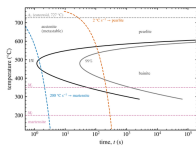
Techniques: shading



71. Ellingham diagram

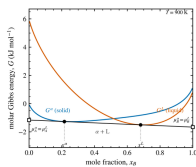
Ellingham diagram from $\Delta H - T\Delta S$ with the Mg boiling-point kink, the exact 1000 K carbothermic Fe crossing, along-line labels plus nudged right-margin labels with leaders

Techniques: direct labels



72. TTT diagram

TTT diagram whose C-curves emerge from $\ln t = a/T + b/(T(T_m - T)^2)$, with A_1 /Ms/Mf lines and two cooling paths (quench $\rightarrow$ martensite, slow $\rightarrow$ pearlite)

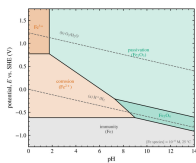


73. Common-tangent construction

Common-tangent construction on regular-solution G curves, solved robustly by parameterising the tangent by slope (brentq), with chemical-potential intercepts and coexisting compositions

Techniques: root finding, tangency construction

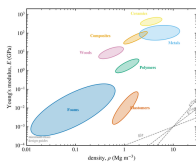
F • Electrochemistry, stability and selection (figs. 74–79)



74. Pourbaix diagram

Simplified Fe Pourbaix (E–pH) diagram: every boundary a Nernst line, self-consistent triple points, water-stability window, filled immunity/corrosion/passivation regions

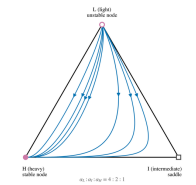
Techniques: shading



75. Ashby chart: E–rho

Ashby modulus–density selection chart: material-family ellipses drawn parametrically in log space, minimum-mass guide lines E/ρ , $E^{1/2}/\rho$, $E^{1/3}/\rho$ with aspect-matched labels

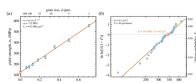
Techniques: direct labels, aspect-aware rotation



76. Residue curve map

Ternary residue curve map integrated from $dx/d\xi = x - y$ (constant α), node topology labelled, direction arrows anchored mid-descent to dodge the near-node crawl

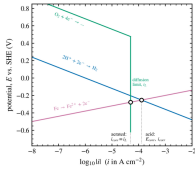
Techniques: direct labels



77. Hall–Petch + Weibull

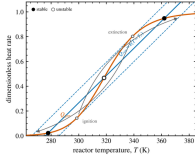
Hall–Petch fit with a nonlinear secondary grain-size axis, plus Weibull median-rank analysis with the double-log linearisation and a secondary failure-probability axis

Techniques: secondary axis



78. Evans diagram

Evans mixed-potential diagram: anodic dissolution vs activation- and diffusion-controlled cathodics; $i_{corr} = i_L$ when the anodic line meets the vertical limit



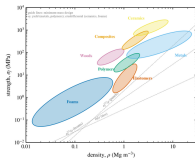
79. CSTR multiplicity

CSTR van Heerden diagram: sigmoid heat generation vs removal lines, three steady states (stable/unstable), ignition and extinction tangencies found by brentq on $x'(T) = \kappa$

Techniques: root finding

G • Ashby charts on the shared module (figs. 80–81)

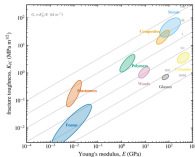
Both charts are thin scripts over `ashby.py`: log-space bubbles with the major axis along the tilt, decade axes, and guide-line labels rotated from the *rendered* axes aspect — the module that retired failure pattern 2.



80. Ashby chart: strength–density

Ashby strength–density chart on the module, with minimum-mass guide lines for ties (σ/ρ), beams ($\sigma^{2/3}/\rho$) and panels ($\sigma^{1/2}/\rho$)

Techniques: shared module

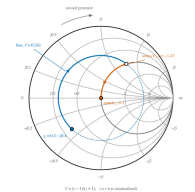


81. Ashby chart: toughness–modulus

Ashby K_{IC} – E chart with toughness contours $G_c = K^2/E$ (slope $\frac{1}{2}$), making the square in the toughness relation visible

Techniques: contours

H • Across science and engineering (figs. 82–100)

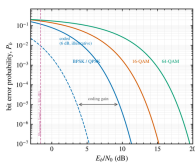


82. Smith chart

RF

Smith chart generated from the bilinear map $\Gamma = (z-1)/(z+1)$, with a worked two-step match: a series line ($\ell \approx 0.24\lambda$, clockwise at constant $-\Gamma$) onto $r = 1$, then a series capacitor to the centre

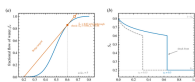
Techniques: series solution



83. BER waterfalls

COMMUNICATIONS

BER waterfalls from Q-function expressions for BPSK/QPSK, 16-QAM and 64-QAM, an illustrative 6 dB coding-gain shift, and the Shannon limit at -1.59 dB

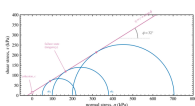


84. Buckley–Leverett waterflood

PETROLEUM

Buckley–Leverett waterflood: Welge tangent from $(S_{wc}, 0)$ fixing the shock saturation and breakthrough average, plus rarefaction-and-shock saturation profiles at two times

Techniques: tangency construction

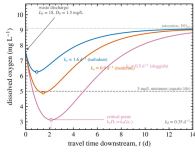


85. Mohr–Coulomb failure

GEOTECHNICAL

Mohr–Coulomb failure circles tangent to $\tau = c + \sigma \tan \phi$ by construction (centre-to-line distance equals radius analytically), with the friction angle and cohesion annotated at equal aspect

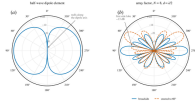
Techniques: annotation, tangency construction, aspect-aware rotation



86. Streeter–Phelps DO sag

ENVIRONMENTAL

Streeter–Phelps dissolved-oxygen sag for three reaeration rates, critical points from the analytic t_c and D_c , and the 5 mg/L aquatic-life threshold

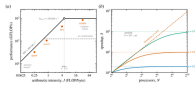


87. Antenna patterns

ANTENNAS

Polar dB patterns computed from array theory: half-wave dipole element factor and the $N = 8$, $d = \lambda/2$ array factor, broadside and steered to 60 ($\beta = -\pi \cos 60$), with the -13 dB first side lobe

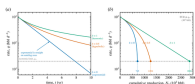
Techniques: polar axes



88. Roofline + Amdahl

COMPUTER ARCH.

Roofline model ($P = \min(P_{peak}, BW \cdot I)$), ridge point, no-vectorisation ceiling, kernels under their roofs; reuses `ashby.setup/slope_angle` beside Amdahl saturation vs Gustafson scaling

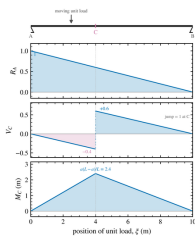


89. Arps decline curves

PETROLEUM

Arps decline family $q = q_i / (1 + b D_{it})^{1/b}$ on semilog time ($b = 0$ straight) and as rate–cumulative with analytic $N_p(q)$: EUR at the economic limit spans $715 \rightarrow 2856 \cdot 10^3$ bbl from $b = 0$ to 1

Techniques: log axes

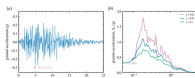


90. Influence lines

STRUCTURAL

Influence lines for R_A , V_C and M_C of a simply supported beam from statics, with beam sketch, positive/negative shading, the unit shear jump at C and the $a(L-a)/L$ moment peak

Techniques: shading

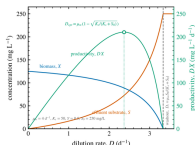


91. Response spectrum

EARTHQUAKE

Elastic response spectrum computed honestly: seeded band-passed, enveloped ground motion (PGA 0.30 g) integrated per-period with a vectorised central-difference SDOF sweep at $\zeta = 2, 5, 10$ %; $S_a \rightarrow$ PGA at short period

Techniques: seeded RNG, envelopes

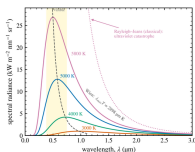


92. Chemostat washout

BIOPROCESS

Chemostat washout diagram from Monod kinetics: S^* , X^* , and productivity $D \cdot X$ vs dilution rate, washout at $D_c = \mu_m S_0 / (K_s + S_0)$, analytic D_{opt} asserted against the numerical maximum

Techniques: built-in self-check

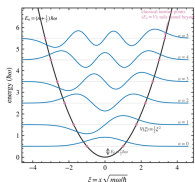


93. Planck blackbody

PHYSICS

Planck spectra at four temperatures with the Wien displacement locus drawn through the peaks (a built-in check), the Rayleigh–Jeans divergence, and the visible band

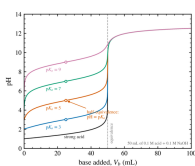
Techniques: built-in self-check



94. Quantum harmonic oscillator

QUANTUM

Quantum harmonic oscillator: Hermite–Gaussian eigenfunctions riding their levels $E_n = (n + \frac{1}{2})\hbar\omega$ inside $V = \xi^2/2$, classical turning points dotted, zero-point energy bracketed at the well bottom

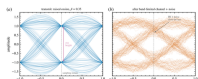


95. Titration curves

CHEMISTRY

Titration curves solved exactly from the proton condition with `brentq` (dilution included): strong + four weak acids, half-equivalence circles landing at pH = pKa by construction

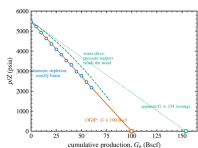
Techniques: root finding



96. Eye diagrams

COMMUNICATIONS

Eye diagrams from folded raised-cosine traces ($\beta = 0.35$, singularities handled analytically): the Nyquist pinch at ± 1 at the transmitter, then ISI + noise closing the eye after a band-limited channel

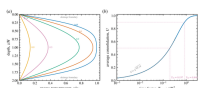


97. p/Z material balance

PETROLEUM

Gas material balance: noisy synthetic surveys on the exact p/Z line recover OGIP by extrapolation ($G_{hat} = 99.8$ vs true 100), with the water-drive bend and its misleading early-time tangent (apparent $G \approx 154$)

Techniques: tangency construction

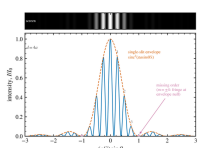


98. Terzaghi consolidation

GEOTECHNICAL

Terzaghi consolidation from the diffusion series: pore-pressure isochrones for a doubly drained layer and $U(T_v)$ with the $\sqrt{4T_v/\pi}$ approximation and the $T_{.50}/T_{.90}$ milestones

Techniques: series solution

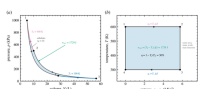


99. Double-slit interference

OPTICS

Double-slit interference under its single-slit envelope, with a rendered screen strip: $d = 4a$ makes the $m = \pm 4$ orders coincide with envelope nulls and vanish — the missing orders are the built-in check

Techniques: built-in self-check, envelopes



100. Carnot cycle, both planes

THERMODYNAMICS

The Carnot cycle in both planes: computed p-V branches enclosing the net work beside the T-s rectangle, with the loop integral asserted equal to $(T_h - T_c)\Delta S$ — 1729 J either way, because work between the same states through the same reversible cycle cannot care which plane you drew it in

Techniques: built-in self-check

3 The failure-mode catalog

Sixteen named patterns, every one harvested from a real bug in building this library. Each surfaced as a wrong *picture* (or a crash) from plausible code; the render-and-inspect loop caught them all.

Escape sequences in raw strings. `\u00b5` inside `r"..."` prints itself. Type the literal character. (Recurred seven times before it became reflex.)

Aspect-blind slope labels. A label rotated by $\arctan(\text{slope})$ ignores the axes' unit or decade aspect and slashes across its own line. Compute the visual angle from the rendered axes geometry (figs. 61, 69, 78; automated in `ashby.py`).

Insets over data. An inset that covers curves must move or become a panel (figs. 53, 63).

Legend-curve collisions. Legends are placed blind; check the swatch column against every curve, or label directly (fig. 70).

Nonlinear secondary axes and minor ticks. A $1/s^2$ transform asks the minor locator to enumerate ticks to 10^{18} ; disable minor locators on nonlinear secondary axes (fig. 77).

Stiff ODEs need terminal events. Boundary-layer shooting blows up past the far boundary; add terminal events instead of praying (fig. 63).

One sign, many symptoms. A single sign slip in one crossing formula truncated a boundary, spawned two stray segments and a bow-tie polygon — four symptoms, one cause (fig. 74). Multiple anomalies usually share a root.

Positional spill into a keyword. `f(ax, *t[1:], label=t[0])` silently feeds a trailing positional into `label`. Unpack tuples explicitly. (Recurring verbatim: figs. 75, 80, 81.)

Fixed-point crawl breaks index arrows. Trajectories crawl near equilibria, so index-based arrow placement collapses to zero length and renders with arbitrary orientation; anchor arrows by *value* and give them a minimum spatial length (fig. 76).

Bottom labels vs. the tick row. Symbolic labels under the axis land on the tick numbers; put them inside the axes (figs. 84, 85).

Converging curves pile their labels. Curves that meet at an edge stack right-edge labels into mush; label each curve where the family is spread (figs. 86, 95).

Guide-line crossover corridors. Two guide lines through different anchors swap vertical order at their crossing; a label corridor that works on one side fails on the other (fig. 80).

Helper contracts are nonlocal. Calling a log–log helper before the axes are logarithmic fed $\log_{10}$ a linear autoscale range, produced NaN, and crashed *inside the font renderer* — errors surface far from their cause (fig. 88).

The renderer speaks a dialect. `\tfrac` is fluent ams $\text{\LaTeX}$ and a fatal error in `mathtext` (fig. 94). Habits are not portable across renderers.

Arrows point at coordinates, not meanings. The Rayleigh–Jeans annotation landed on the Planck peak instead of the classical curve it named; aim at the named object’s ink (fig. 93).

Algebraic label collisions. Sometimes the collision is a theorem: a label offset exactly equal to the Henderson–Hasselbalch term put every titration label on its own curve (fig. 95); a productivity curve passed through the biomass label because $DX = X$ at $D = 1$ there (fig. 92). When a label lands on a curve at one x , suspect the algebra, then move along the curve.

4 Checklists

Before writing

- Identify the governing equations and the landmark values the figure must honour; plan the built-in self-check.
- Choose column width, panel count and message-per-panel; pick which curves get direct labels and where the family is separated.

Before rendering

- Style sheet loaded; sizes in inches; symbols as literal characters; axis labels carry units; RNG seeded.
- Rotated labels use rendered-aspect angles; secondary nonlinear axes have minor locators disabled; tuples unpacked explicitly.

Inspecting the render

- Every label: on its object, off the data, inside the frame, clear of tick rows and of every other label?
- Every arrow: finite length, pointing at the named ink, correct direction along the curve?
- Landmarks and self-checks: do computed values match the physics (peaks on loci, tangencies exact, areas equal, limits approached)?
- Anything clipped at a spine? Anything only visible because you zoomed? Fix, rerun, look again.

Shipping

- PNG + vector PDF written; confirmation printed with key numbers.
- README line added; library zip rebuilt; script runs standalone from a clean directory.

Closing note. The library's real product is not the hundred pictures but the loop that made them: derive, render, inspect, fix, and let the physics check itself. Everything else in this guide is that loop, written down.